

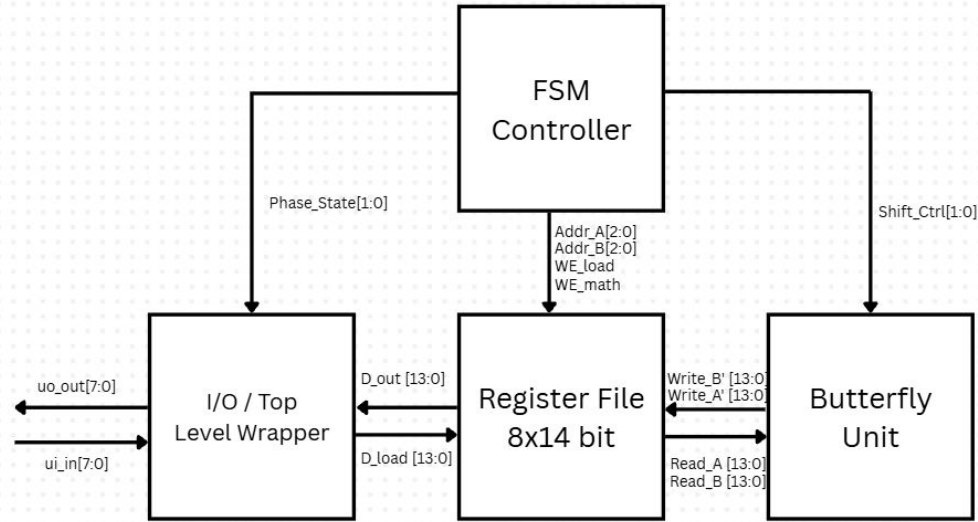
8-Point FFT Processor

Fast Fourier Transform: An algorithm that converts a signal from the time domain into the frequency domain

What does 8-Point Mean? Takes 8 discrete real-valued time-domain samples and produces 8 complex-valued frequency bins

Each input is a 7-bit real number, and each output is a 14-bit complex number

System Diagram



I/O pin assignment table (Tiny Tapeout pins)

Name	Count	Direction	Description
clk	1	Input	Clock input
rst_n	1	Input	Active low reset
ui_in[7:0]	8	Input	8 bit input in time domain
uo_out[7:0]	8	Output	8 bit output in frequency domain
uio[0]	1	Bidir	is_valid_input
uio[1]	1	Bidir	is_valid_output

Specifications

Architecture Type: Radix-2 Decimation-In-Time (DIT), Folded Serial Datapath

Internal Word Size: 14 bits total per sample.

Real Component: 7 bits (1 sign bit, 6 data bits, two's complement).

Imaginary Component: 7 bits (1 sign bit, 6 data bits, two's complement).

Input Format: 8-bit time-domain samples on the `ui_in[7:0]` bus

Timeline + Dividing Work

What	Due Date	Who Does
Create register file	Oct 15	Chris
Create FSM controller	Oct 15	Daniel
Create Butterfly Unit	Nov 1	Chris
Create I/O (Top-level wrapper)	Nov 1	Daniel
Integration	Nov 15	Both
Python CocoTB testing	Nov 30	Both